INCH-POUND

A-A-52566

May 23, 1997

COMMERCIAL ITEM DESCRIPTION

SINKS, PHOTOGRAPHIC PROCESSING, TEMPERATURE-CONTROLLED

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-S-13565(ME), for all federal agencies.

1. SCOPE.

1.1 This commercial item description covers photographic processing sinks (hereinafter referred to as sink(s)) consisting of a base-mounted, insulated trough; a water-circulation system with a temperature control unit, that shall be as described herein (see figure 1). The sinks are intended for use in truck-mounted and base-installation photographic laboratories. Type I and type II, size 1 sinks are to be used in truck-mounted darkrooms; type II sizes 1 and 2 are to be used in base-installations.

2. CLASSIFICATION. The sinks shall conform to the following types and sizes.

2.1 Type I.

Size 1 - Accommodates three 9 by 14 inch (23 by 36 cm), outside dimension, straight-sided trays.

2.2 Type II.

Size 1 - Accommodates three 20 by 24 inch (51 by 61 cm) capacity trays.

Size 2 - Accommodates three 30 by 40 inch (76 by 102 cm) capacity trays.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to:
U.S. ARMY TOPOGRAPHIC ENGINEERING CENTER, ATTN: CETEC-PD-TE,
7701 TELEGRAPH ROAD, ALEXANDRIA, VA 22315-3864.

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3. SALIENT CHARACTERISTICS.

3.1 Base. The base of the sinks shall be constructed of materials specified herein and shall have removable panels for access to the temperature control unit(s). A means of leveling the sink shall be provided in the base. The sink shall not twist, buckle or warp under any of the operating or transport conditions specified herein. The type I sinks shall withstand a sustained, evenly distributed weight of 500 lb (227 kg); the type II sinks, a sustained, evenly distributed weight of 1000 pounds (454 kg). All sinks shall withstand a horizontal force of 100 lb (45 kg).

3.2 Trough. The trough shall overhang the front and sides of the base by 3.00 in. (8 cm) and shall be supported and braced to prevent buckling and sagging during travel and in normal use. The trough shall have double walls sandwiching commercial grade, fungus and vermin resistant insulation. The inner trough shall form a basin to contain a constant temperature, circulating water bath and shall be provided with a drain. The inlet and outlet for the water-circulating system shall be provided with strainers and located to ensure maximum circulation and uniform temperature of the water throughout the basin. The bottom of the basin shall have rounded, upward protrusions not less than .25 in. (.64 cm) high, not less than 1.50 in. (4 cm) in diameter, and spaced not greater than 8.00 in. (20 cm) center-to-center. The protrusions shall show no stretching or fatigue and the trough shall not leak under any of the operating or transport conditions specified herein.

3.2.1 Dump gutter. A dump gutter or channel, when provided, shall not encroach on the inside dimensions of the basin and shall not drain into the water-circulating system.

3.2.2 Overflow outlet. An overflow outlet, emptying directly into the trough drain, shall be provided to prevent the water in the basin from reaching a depth greater than 2.50 in. (6 cm).

3.3 Temperature controlled, water-circulating system. The temperature controlled, water-circulating system shall be driven by an electric pump. The components of the pump shall be corrosion-resistant and the motor shall have sealed bearings. If the pump is located at other than the lowest point in the water-circulating system (except for the drain), an integral, self-priming pump shall be used. The type I sink may be supplied without the pump and circulating system, if it otherwise meets all the temperature control requirements specified herein.

3.3.1 Plumbing. Piping and fittings shall not leak and shall not introduce or permit entry of contaminants into the water-circulating system. All piping shall be insulated, except where temperature dissipation is required. Unions shall be used at joints in the water lines, to facilitate disassembly, where necessary for service or replacement of components. Evaporative coils, in contact with the circulating water, shall be one continuous length. The drain for type I sinks shall not be less than 1 inch iron pipe size (IPS); the type II sinks, not less than 1-¼ inch IPS.

3.4 Temperature-control unit. The temperature-control unit shall consist of an electric water-heating element, a refrigeration unit, and the necessary thermostatic controls to regulate the temperature of the water. The controls shall maintain the water temperature at 68° F (20° C) ± 1.0° F, throughout an ambient temperature range of 45° F (7° C) to 110° F (43° C).

3.4.1 Heating element. The trough water shall be heated by an immersion-type, electric, heating element. The element(s) shall be capable of heating the liquid in the trays from 45° (7° C) F to 68° F (20° C) in one hour or less.

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3.4.2 Refrigeration unit. The refrigeration unit shall consist of an electric-motor-driven, sealed compressor; a condenser; thermostatically-controlled solenoid valves; refrigerant expansion coils with an adjustable-type expansion valve; a disposable drier-filter; and a sight glass. The unit shall be charged with a non-toxic, environmentally safe refrigerant. Seamless tubing and solder-type fittings shall be used and the subsystem shall incorporate high- and low-pressure gages.

3.4.2.1 Condenser. Unless otherwise specified (see 6.1), the refrigeration unit condenser shall be cooled by an air-circulating fan. The condenser air outlet shall be designed for the attachment of exhaust air ducts, at the rear of the unit (see figure 2), that shall be readily accessible.

3.4.2.2 Cooling capacity. The refrigeration unit for the sink shall be capable of cooling the liquid in the trays from 80° F (27° C) to 68° F (20° C), in an ambient temperature of 110° F (43° C). The time required to reduce and stabilize the temperature shall not be greater than 1 hour for the type I, size 1 sink; not greater than 2 hours for the type II, size 1 sink; and not greater than 3 hours for the type II, size 2 sink.

3.5 Storage cabinet. Each type II sink shall include a storage cabinet in the base. The storage area of the cabinet shall be temperature controlled to maintain a constant 68° F (20° C) ± 1.0° F, using the sub-system installed for the water-circulating, constant temperature bath. Control of the storage cabinet temperature shall be automatic and independent of other components of the sink, being controlled by a separate thermostat, solenoid valve, and adjustable-type expansion valve. The cabinet shall be insulated on all six sides with a non-settling, fungus- and vermin-resistant insulation. The door shall open to expose the full height and width of the cabinet and shall have a durable, resilient, air tight seal. The interior of the cabinet shall be constructed of a material that is inherently resistant to the effects of corrosive chemical agents, normally used in photographic processing.

3.5.1 Rack. The cabinet shall contain a rigid, wire-frame rack, of a similar corrosion-resistant material, constructed to retain and support the required number of solution bottles. The size 1 sink shall have a capacity for not less than six 1.00 gallon (3.785 liter) solution bottles; the size 2, not less than eight. The bottles are 5.66 in. (14.38 cm) in diameter and 13.56 in. (34.44 cm) in height. The rack shall slide out for unobstructed access to the bottles and shall permit their easy insertion and removal.

3.5.2 Drip pan. The cabinet shall be provided with a drip pan. The drip pan shall fit under the condenser coils and shall be easily removable without tipping.

3.6 Control switch.

3.6.1 Type I. Type I sinks shall be provided with a two-position switch marked “ON” and “OFF”. The markings shall be on or adjacent to the switch. When the switch is set in the “ON” position, the temperature control functions shall operate.

3.6.2 Type II. Type II sinks shall have a three-position switch with plate or dial marked “OFF”, “CABINET” and “SINK”. When the switch is set in the “CABINET” position, the temperature control functions of the storage cabinet shall operate. When the switch is set in the “SINK” position, the temperature control functions for the bath and storage cabinet, and the water circulation system shall operate.

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3.7 Electrical requirements. The electrical components of the sinks shall operate on either a 120 vac, 60 Hz, single phase power source or a 208 vac, 60 Hz, 3 phase, 4 wire system (see 6.1). Motors shall conform to NEMA MGI Motors and Generators Standards (see 6.4.1) and shall be provided with thermal-magnetic, inverse time, overload circuit breakers. Wiring shall conform to the National Electric Code, NFPA No. 70 (see 6.4.2). All electrical components and accessories shall be wired to a terminal box containing the terminals for the power line lead.

3.8 Installation requirements. All fixtures, connections and plumbing, exterior to the base of the sink, shall be contained within the envelope defined by the overhang of the trough.

3.8.1 Type I sinks. Floor space requirements for type I, size 1 sinks shall not be greater than 36.00 in. (91 cm) by 34.00 in. (86 cm). The working height shall be 34.00 in. (86 cm) to 38.00 in. (97 cm).

3.8.2 Type II sinks. Floor space requirements for the type II, size 1 sinks shall not be greater than 81.00 in. (206 cm) by 34.00 in. (86 cm); for type II, size 2 sinks, not greater than 113.00 in. (287 cm) by 52.00 in. (132 cm). The working height of both type II sinks shall be 36.00 in. (91 cm) to 38.00 in. (97 cm).

3.9 Transportability. The sink shall be of rigid construction and shall withstand the shock and vibration of road travel, including off-the-road (rough terrain) conditions. When specified (see 6.1) the contractor shall devise a road test, subject to the approval of the contracting officer, to determine the adequacy of the unit's transportability.

3.10 Ease of maintenance. Components requiring adjustment, maintenance, repair, or replacement shall be readily accessible, without requiring removal of major components or assemblies. Covers or plates that must be removed for these purposes shall be provided with quick-disconnected fasteners.

3.11 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable (see 6.5.2).

3.12 Safety. All exposed parts that conduct electricity or that are subject to operating temperature extremes shall be insulated, fully enclosed or guarded. All moving parts presenting a hazard to personnel shall be fully enclosed or guarded. Protective devices shall not impair operating functions.

3.13 Lubrication. All component requiring lubrication shall be lubricated prior to delivery. Fittings shall be self-sealing and located to facilitate access, using standard lubricating equipment.

3.14 Material. Material shall be as specified herein or selected by the contractor, subject to all the provisions of this specification. Materials selected shall be sufficient to withstand the rigors of daily use in contact with corrosive chemical agents, without suffering the effects of corrosive deterioration.

3.14.1 Material deterioration prevention and control. The sinks shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration to which they are susceptible.

3.14.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion.

3.14.3 Identification of materials and finishes. The contractor shall identify and document the specific material and material finish or treatment of all components and sub-components.

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3.14.4 Fungus resistance. Materials selected for use in the sinks shall be fungus-inert or shall be treated with a commercially accepted, moisture and fungus-proofing compound. This requirement need not apply to components that are, or are within enclosures that are, hermetically sealed.

3.14.5 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements and promotes economically advantageous life cycle costs.

3.15 Treatment and painting . Unless otherwise specified (see 6.1), or unless the material is inherently corrosion resistant, the sinks shall be treated and painted for corrosion protection using the contractors standard practice.

3.16 Instruction plates. Each sink shall be equipped with instruction plates describing procedures to be followed in assembling, operating or servicing the sink. The plates shall include all applicable warnings and cautions, describe any special or important procedures, and may include diagrams. Plates shall be placed in a protected, readily visible area on a non-work surface; shall be permanently attached; and remain legible for the life expectancy of the sink.

3.17 Metric products. Products manufactured to metric units will be considered on an equal basis with those manufactured using inch-pound dimensions, provided they fall within specified tolerances and all other requirements of this Commercial Item Description are met. If a product is manufactured to metric units and exceeds the tolerances specified, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

3.18 Certification. When specified in the contract or purchase order (see 6.1), the certification of commercial products as specified in Section 4, may be accepted in lieu of the examinations specified in 4.3.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Certification of commercial item. The contractor shall certify and maintain substantiating evidence that the sinks offered meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards and quality assurance practices, and are the same sinks offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.2 Market acceptability. The company producing the temperature controlled, photographic processing sinks must have been producing sinks meeting the requirements of this CID for at least two years.

4.3 Examinations. Each sink examined, shall be inspected for compliance with the requirements specified in 3.1 through 3.16. Each element of inspection shall encompass all visual and functional examinations. Nonconformance to any specified requirement or the presence of one or more defects, lessening required efficiency, shall constitute cause for rejection. Any redesign or modification following failure to meet the specified requirements shall receive particular attention for adequacy and suitability.

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5. PACKAGING.

5.1 Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. When the condenser shall be other than air cooled (see 3.4.2.1).

c. Whether the sinks shall operate on 120 or 208 volts (see 3.7).

d. When treatment and painting shall be other than the standard practice. (see 3.15).

e. When certification may be accepted in lieu of examination (see 3.16).

f. When road test is required (see 3.9).

6.2 Part Identification Number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

This example describes a part numbering system for CID A-A-52566

AA52566-2-1

- Size 1
- Size 2 (type 2 only)
- Type I
- Type II

6.3 Figures. The figures show limiting dimensions for Type II sinks, and depict a configuration that has been found acceptable. However, they are illustrative only and not intended to preclude the furnishing of another sink that conforms to the requirements of this specification.

6.4 Source of documents.

6.4.1 NEMA MGI - Motors and Generators Standards are available from the National Electrical Manufacturers Association, 155 East 44th Street, New York, NY 10017.

6.4.2 The National Electrical Code, NFPA No. 70, is available from the National Fire Protection Association International, 60 Batterymarch Park, Quincy, MA 02269.

6.4.3 The code of Federal Acquisition Regulations (FAR) - Section 23,403 is available from the Superintendent of Documents , U.S. Government Printing Office, Washington, DC 20402.

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6.5 Definitions. For the purposes of this specification, the following definitions apply.

6.5.1 Fungus-inert. A fungus inert material is a material that, in all modified states and grades, is not a nutrient to fungi.

6.5.2 Interchangeable parts. Interchangeable parts are defined as two or more parts possessing such functional and physical characteristics as to be equivalent in performance and durability. These parts are capable of being exchanged one for the other without alteration of the parts; of mating or adjoining parts, except for adjustment; and without selection for fit or performance.

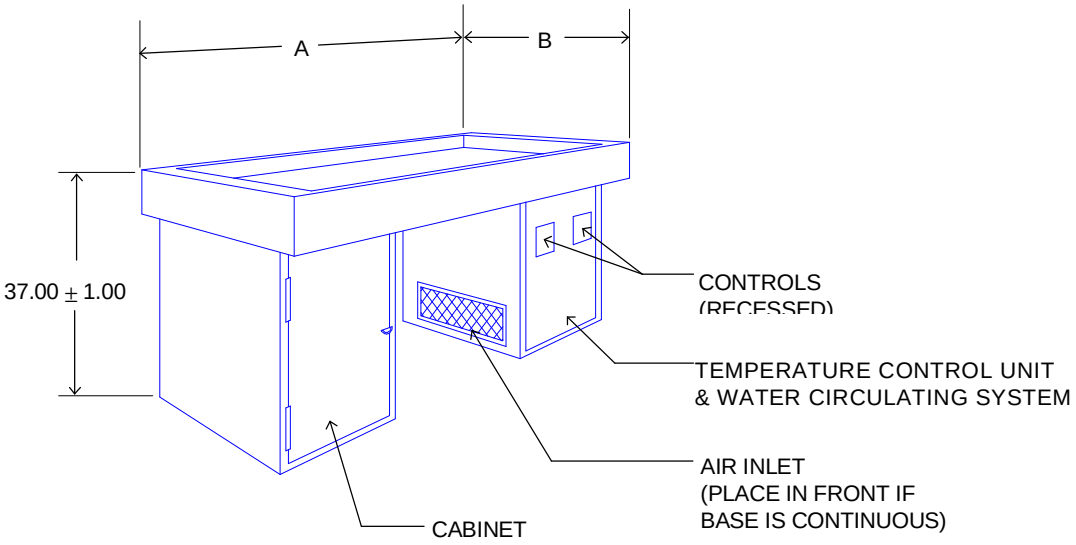
6.6 Data requirements. The contracting officer should include, in the contract or order, any data requirements, such as maintenance and operation manuals, instructional materials and technical publications to be furnished with each sink.

6.7 Provisioning. The contracting officer should include, in the contract or order, provisioning requirements for repair parts; maintenance tools (including any special tools), as necessary; and instructions regarding shipment of the sinks.

6.9 Subject term (key words) listing.

Water-circulation
Constant-temperature
Heating
Cooling
Trough
Cabinet

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NOTE :

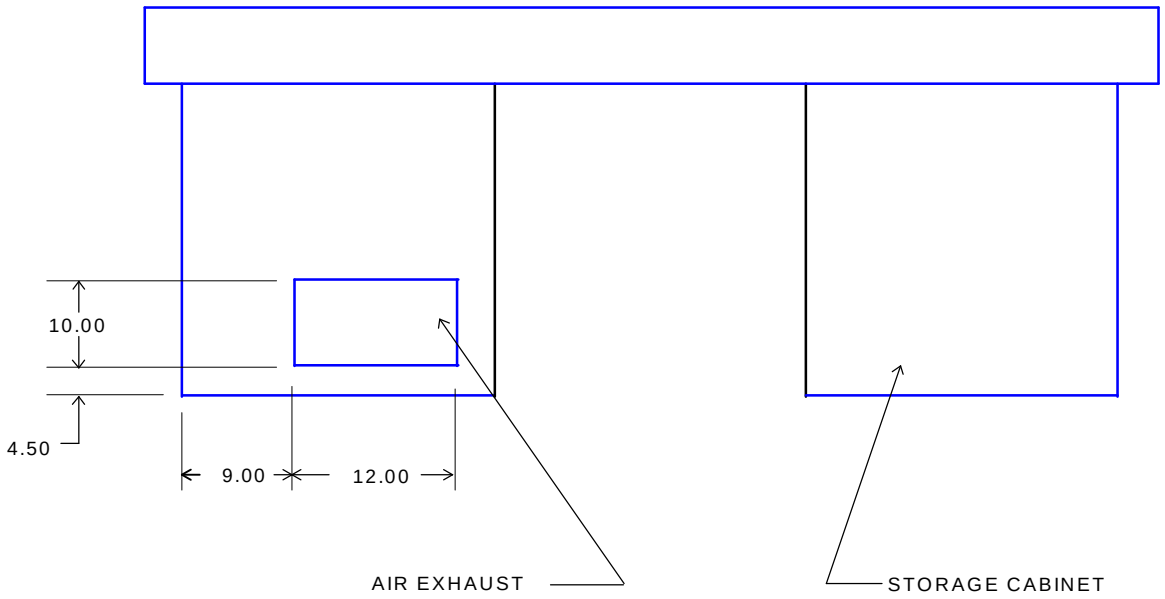
DIMENSIONS ARE IN INCHES.

SIZE	OVERALL DIMENSIONS		INSIDE TROUGH DIMENSIONS (No internal obstructions)		
	A	B	LENGTH	WIDTH	DEPTH
1	80.00 +0.00 -1.00	33.50 ±1.50	76.50 ±.50	30.00 +0.00 -2.00	5.00 ±1.50
2	110.00 ±3.00	50.00 ±2.00	106.00 ±2.00	46.00 ±1.00	5.00 ±1.50

Figure 1. Type II, Temperature Controlled Photographic Sinks.

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REAR VIEW OF TYPE II, SIZE 1 SINK



NOTE :

DIMENSIONS ARE IN INCHES
TOLERANCES ± .12

Figure 2. Condenser exhaust port location (for truck mounting).

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MILITARY INTEREST:

Custodian:
Army - CE

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - FSS

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Project 6740-0016)

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